

Abdulmajid Awol Seid

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Summary

PhD student in Computer Science at Tulane University and graduate research assistant in the Laboratory for Software Design, advised by Dr. Hridesh Rajan. I research reliable and secure AI agents at the intersection of artificial intelligence, software engineering, and programming languages. My current work focuses on preventing compositional and decompositional attacks in AI agents.

Research Interests

Reliable AI agents: Agentic systems, tool use, failure analysis, evaluation
Agent safety and security: Compositional and decompositional attacks, tool misuse, provenance
AI + software engineering: AI4SE, agentic software engineering, empirical software engineering
PL for AI agents: Contracts, state, refinement, provenance, specification

Education

PhD	Tulane University , Computer Science - School of Science and Engineering. Advisor: Dr. Hridesh Rajan. - Research focus: AI/LLM reliability, AI agent safety, and software engineering.	New Orleans, LA, USA Aug 2026 – present
MSc	Addis Ababa Science and Technology University , Software Engineering (Fast Track Program) - Thesis: <i>Predicting High-Risk Technical Debt in Open-Source Software Projects Using Machine Learning</i>. Advisor: Dr. Tesfaye Gidey. - Scholarship for academic excellence to begin the master's before completing the bachelor's degree.	Addis Ababa, Ethiopia Jan 2024 – July 2026
BSc	Addis Ababa Science and Technology University , Software Engineering Graduated with Honors (Very Great Distinction), GPA 3.78/4.0.	Addis Ababa, Ethiopia May 2021 – June 2025
	Africa to Silicon Valley (A2SV) , Software Engineering and Algorithms Training 1,000+ hours of data structures and algorithms training; 600+ problems solved on LeetCode and Codeforces.	Addis Ababa, Ethiopia Nov 2023 – Nov 2024

Thesis

Predicting High-Risk Technical Debt in Open-Source Software Projects Using Machine Learning Abdulmajid Awol Seid github.com/maajidAwol/technical-debt (Master's thesis, Addis Ababa Science and Technology University)	July 2026
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Research Experience

Tulane University, Laboratory for Software Design , Graduate Research Assistant - Investigating how to prevent compositional and decompositional attacks in tool-using AI agents. - Exploring software engineering and programming-language techniques (contracts, state, provenance) for reasoning about agent behavior.	New Orleans, LA, USA Aug 2026 – present
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Addis Ababa Science and Technology University, MSc Thesis Research

- Built a reproducible cross-project framework that predicts which files become maintenance-intensive (27 features, 22 Apache Java projects). On unseen projects, the top 20% of files ranked by the best model recover about 82% of future maintenance-intensive files, about 4.1x random inspection.

Addis Ababa, Ethiopia

Completed July 2026

Engineering Experience

Ethiopian Artificial Intelligence Institute, Software Engineer

Addis Ababa, Ethiopia

Aug 2025 – June 2026

Upwork (Freelance), including Omnicasa, Brussels, Full Stack Developer

Remote

- Built a distributed video processing pipeline with Redis-backed job queues and asynchronous workflows, cutting turnaround from 2 hours to 15 minutes per recording.
- Automated transcription and post-meeting analytics, reducing review time by 30%. Top Rated freelancer.

July 2024 – Mar 2026

iCog Labs, AI/ML Intern

Addis Ababa, Ethiopia

May 2025 – Aug 2025

AASTU Software Engineers Association, Software Engineer Associate

Addis Ababa, Ethiopia

Sept 2024 – June 2025

Eskalate LLC, Mobile Application Developer

Addis Ababa, Ethiopia

- Built Flutter mobile applications in a team following test-driven development.

June 2024 – Nov 2024

Camara Education Ethiopia, Full Stack Developer Intern

Addis Ababa, Ethiopia

- Contributed to an educational platform used by 200+ Ethiopian schools and 300,000+ students.

July 2023 – Nov 2023

Selected Projects

Miiltoo Knowledge Companion

Aug 2023 – Nov 2023

- Led a team of five building a generative-AI chatbot and quiz bot for Ethiopian Grade 12 exam preparation (Flask, MySQL, OpenAI API).
- Top 9 of 3,700+ participants from 47 countries, A2SV Generative AI for Africa Hackathon.

Flutter Clean Architecture VS Code Extension

Dec 2024 – Mar 2025

- Open-source extension (190+ installs) that scaffolds and auto-repairs Clean Architecture structure in Flutter apps.

Honors and Awards

2025: Academic Excellence Award, Graduated with Honors (Very Great Distinction), AASTU**2024:** MSc Fast Track Scholarship for Academic Excellence, AASTU**2023:** Finalist (Top 9 of 3,700+), A2SV Generative AI Hackathon for Africa**2021:** University Presidential Award, AASTU

Skills

Programming: Python, Java, C++, JavaScript, TypeScript, Dart, PHP, SQL**AI / ML:** Supervised learning (logistic regression, random forests, XGBoost, LightGBM), model evaluation, LLM APIs, RAG**Software engineering:** Software architecture, testing, distributed and asynchronous systems, CI/CD, Git, Docker**Research methods:** Empirical software engineering, ML for software engineering, cross-project validation, reproducible pipelines